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TAWHIDIC EPISTEMOLOGY LEADING THE WAY UMMATIC EXCELLENCE LEADING THE WORLD
KHALĪFAH • AMĀNAH • IQRA' • RAḤMATAN LIL-ĀLAMĪN

BICS 2304

COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

PROJECT TITLE

WEARABLE HEAT STROKE & DEHYDRATION PREVENTION SYSTEM

Submission for Hardware Listings and Description

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SECTION: 04

LECTURER: YASIR MEHMOOD

PREPARED BY:

NO.	NAME	MATRIC NO.	ROLE
1	ADAM FAWWAZ BIN SAZALIZAM	2416969	Member
2	AHMAD FAWWAZ BIN ABDUL KADIR	2418497	Member
3	WAN ABDULLAH MIKHAIEL BIN WAN SHAMSUDDIN	2416229	Member
4	ALFYAN SHAWN BIN JASNI SHIRLIN	2417841	Member

Hardware List and Description

The following hardware components are proposed for the implementation of the **Wearable Heat Stroke & Dehydration Prevention System**. These components are selected based on the system requirements, sensor-actuator logic, and real-world feasibility for wearable IoT applications in tropical environments such as Malaysia.

No.	Component Category	Hardware Component	Exact Model	Specifications / Operating Range	Function in System
1	Microcontroller	ESP32 DevKit V1	ESP32-WROOM-32	Dual-core 240 MHz processor, 520 KB SRAM, Wi-Fi + Bluetooth, operating voltage 3.3V	Main embedded controller that processes sensor data and controls actuators
2	Temperature Sensor	MAX30205 Human Body Temperature Sensor	MAX30205	Temperature range: 0°C–50°C, accuracy ±0.1°C between 37°C–39°C, I2C communication	Measures the wearer's body temperature for heat stress analysis
3	Pulse Rate Sensor	MAX30102 Pulse Oximeter and Heart Rate Sensor	MAX30102	Heart rate range: 30–250 BPM, low-power operation, I2C interface	Detects heart rate and physical exertion level
4	Motion Sensor	MPU6050 Accelerometer and Gyroscope Module	MPU6050	3-axis accelerometer and gyroscope, acceleration range ±2g to ±16g	Detects movement, orientation, sudden deceleration and collapse
5	Haptic Actuator	Mini Coin Vibration Motor 3V	1027 Coin Vibration Motor	Operating voltage 2.5V–3.5V, vibration speed ~10,000 RPM	Produces vibration alerts during heat stress or hydration reminders
6	Visual Actuator	5mm High Brightness Red LED	Generic 5mm LED	Forward voltage 2V, operating current 20mA	Provides hydration reminder and warning indication

7	Cooling Actuator	Mini DC Cooling Fan 5V	5V Brushless Fan	Operating voltage 5V DC, low power consumption	Activates cooling assistance during critical heat stress
8	GPS Module	NEO-6M GPS Module	NEO-6M	Position accuracy 2.5 meters, UART communication, 3V–5V operation	Provides location coordinates during emergency situations
9	Communication Module	SIM800L GSM Module	SIM800L	Quad-band GSM/GPRS, operating voltage 3.4V–4.4V	Sends SOS notifications to emergency contacts
10	Display Module	SSD1306 0.96-inch OLED Display	SSD1306 OLED	128×64 resolution, I2C communication, low power display	Displays system safety status and alerts
11	Power Supply	18650 Li-ion Rechargeable Battery	18650 Li-ion Battery	3.7V nominal voltage, 2200–3000mAh capacity	Portable rechargeable power source for wearable operation
12	Voltage Regulation	LM2596 DC-DC Buck Converter	LM2596	Input 4V–40V, adjustable output 1.25V–35V	Regulates stable voltage supply to all modules

Hardware Mapping to Project Logic

The hardware components directly support the behaviour specified in the project documentation:

- The temperature sensor and pulse sensor work together to calculate the Heat Stress Index condition:
  - Temperature > 38°C
  - Heart Rate > 130 BPM
- The accelerometer detects sudden falls and zero-motion conditions.
- The vibration motor, LED, and cooling fan provide immediate user intervention feedback.
- The GPS and GSM modules support the emergency SOS functionality.
- The OLED display outputs user safety status such as:
  - Status: Safe
  - Status: Take Break
  - Drink Water
  - Status : CRITICAL